

*There is grandeur in this view of life, with its several powers, having been originally breathed into a few forms or into one; and that, whilst this planet has gone cycling on according to the fixed law of gravity, from so simple a beginning endless forms most beautiful and most wonderful have been, and are being, evolved.*

# 1

## Customisations and extensions

This chapter describes a number of additional tips and tricks as well as possible customizations to the `oxforddown` thesis.

### 1.1 Chunk caching and the `__bookdown_files` folder

If you set `cache=TRUE` in a code chunk, in order to cache its results if it's time-consuming to run see the R Markdown documentation, then the files for the caching are stored in the `**__bookdown_files**` folder.

If you don't use caching and you would like to just have the `**__bookdown_files**` folder deleted after the build process is complete, then set `allow_cache = FALSE` in `index.Rmd`'s call to `knit_thesis`.

That is, your YAML should then look like this:

```
knit: (function(input, ...) {  
  thesis_formats <- "pdf";  
  
  source("scripts_and_filters/knit-functions.R");  
  knit_thesis(input, thesis_formats, allow_cache = FALSE, ...)  
})
```

## 1.2 Front matter

### 1.2.1 Shorten captions shown in the list of figures (PDF)

You might want your list of figures (which follows the table of contents) to have shorter (or just different) figure descriptions than the actual figure captions.

Do this using the chunk option `fig.scap` ('short caption'), for example `{r captain-image, fig.cap="A very long and descriptive (and potentially boring) caption that doesn't fit in the list of figures, but helps the reader understand what the figure communicates.", fig.scap="A concise description for the list of figures"`

### 1.2.2 Shorten captions shown in the list of tables (PDF)

You might want your list of tables (which follows the list of figures in your thesis front matter) to have shorter (or just different) table descriptions than the actual table captions.

If you are using `knitr::kable` to generate a table, you can do this with the argument `caption.short`, e.g.:

```
knitr::kable(mtcars,
  caption = "A very long and descriptive (and
    ↪ potentially
    ↪ boring) caption that doesn't fit in the list of
    ↪ figures,
    ↪ but helps the reader understand what the figure
    ↪ communicates.",
  caption.short = "A concise description for the list of
    ↪ tables")
```

## 1.3 Shorten running header (PDF)

You might want a chapter's running header (i.e. the header showing the title of the current chapter at the top of page) to be shorter (or just different) to the actual chapter title.

Do this by adding the latex command `\chaptermark{My shorter version}` after your chapter title.

For example, chapter ??'s running header is simply 'Cites and cross-refs', because it begins like this:

```
# Citations, cross-references, and collaboration {#cites-and-refs}
\chaptermark{Cites and cross-refs}
```

## 1.4 Unnumbered chapters

To make chapters unnumbered (normally only relevant to the Introduction and/or the Conclusion), follow the chapter header with `{-}`, e.g. `# Introduction {-}`.

When you do this, you must also follow the heading with these two latex commands:

```
\adjustmtc
\markboth{The Name of Your Unnumbered Chapter}{{}}
```

Otherwise the chapter's mini table of contents and the running header will show the previous chapter.

## 1.5 Beginning chapters with quotes (PDF)

The OxThesis LaTeX template lets you inject some wittiness into your thesis by including a block of type `savequote` at the beginning of chapters. To do this, use the syntax ````{block type='savequote'}`.<sup>1</sup>

Add the reference for the quote with the chunk option `quote_author="my author name"`. You will also want to add the chunk option `include=knitr::is_latex_output()` so that quotes are only included in PDF output.

It's not possible to use markdown syntax inside chunk options, so if you want to e.g. italicise a book name in the reference use a 'text reference': Create a named

---

<sup>1</sup>For more on custom block types, see the relevant section in *Authoring Books with R Markdown*.

piece of text with `(ref:label-name) My text`, then point to this in the chunk option with `quote_author='(ref:label-name)'`.

## 1.6 Highlighting corrections (HTML & PDF)

For when it comes time to do corrections, you may want to highlight changes made when you submit a post-viva, corrected copy to your examiners so they can quickly verify you've completed the task. You can do so like this:

### 1.6.1 Short, inline corrections

Highlight **short, inline corrections** by doing `[like this]{.correction}` — the text between the square brackets will then be highlighted in blue in the output.

Note that pandoc might get confused by citations and cross-references inside inline corrections. In particular, it might get confused by `"[what @Shea2014 said]{.correction}"` which becomes `[Shea2014]{.correction}`. In such cases, you can use LaTeX syntax directly. The correction highlighting uses the soul package, so you can do like this:

- If using biblatex for references, use `"\hl{what \textcite{Shea2014} said}"`
- If using natbib for references, use `"\hl{what \cite{Shea2014} said}"`

Using raw LaTeX has the drawback of corrections then not showing up in HTML output at all, but you might only care about correction highlighting in the PDF for your examiners anyway!

### 1.6.2 Blocks of added or changed material

Highlight entire **blocks of added or changed material** by putting them in a block of type `correction`, using the syntax ````\block type='correction'`.2 Like so:`

---

<sup>2</sup>In the `.tex` file for PDF output, this will put the content between `\begin{correction}` and `\end{correction}`; in gitbook output it will be put between `<div class="correction">` and `</div>`.

For larger chunks, like this paragraph or indeed entire figures, you can use the `correction` block type. This environment **highlights paragraph-sized and larger blocks** with the same blue colour.

*Note that correction blocks cannot be included in word output.*

### 1.6.3 Stopping corrections from being highlighted

To turn off correction highlighting, go to the YAML header of `index.Rmd`, then:

- PDF output: set `corrections: false`
- HTML output: remove or comment out - `templates/corrections.css`

## 1.7 Apply custom font color and highlighting to text (HTML & PDF)

The lua filter that adds the functionality to highlight corrections adds two more tricks: you can apply your own choice of colour to highlight text, or change the font color. The syntax is as follows:

Here's `[some text in pink highlighting]{highlight="pink"}`  
Becomes: Here's some text in pink highlighting.

`[Here's some text with blue font]{color="blue"}`  
Becomes: Here's some text with blue font

Finally — never, ever actually do this — `[here's some text with black highlighting and yellow font]{highlight="black" color="yellow"}`  
Becomes: here's some text with black highlighting and yellow font

The file `scripts_and_filters/colour_and_highlight.lua` implements this, if you want to fiddle around with it. It works with both PDF and HTML output.

## 1.8 Adding a second abstract (PDF)

You may need two abstracts in your thesis, if you e.g. need both an abstract in English and some other language.

You can add a second abstract in `index.Rmd` like so:

```
abstract-second-heading: "Résumé"
abstract-second: "This is the second abstract, for example in
↪ beautiful French."
```

## 1.9 Including another paper in your thesis - embed a PDF document

You may want to embed existing PDF documents into the thesis, for example if your department allows a ‘portfolio’ style thesis and you need to include an existing typeset publication as a chapter.

In gitbook output, you can simply use `knitr::include_graphics` and it should include a scrollable (and downloadable) PDF. You will probably want to set the chunk options `out.width='100%'` and `out.height='1000px'`:

```
knitr::include_graphics("figures/sample-content/pdf_embed_example/Lyngs2020_FB.pdf")
```

In LaTeX output, however, this approach can cause odd behaviour. Therefore, when you build your thesis to PDF, split the PDF into an alphanumerically sorted sequence of **single-page** PDF files (you can do this automatically with the package `pdftools`). You can then use the appropriate LaTeX command to insert them, as shown below (for brevity, in the `oxforddown` PDF sample content we’re only including two pages). *Note that the chunk option `results='asis'` must be set.* You may also want to remove margins from the PDF files, which you can do with Adobe Acrobat (paid version) and likely other software.

```
# install.packages(pdftools)
# split PDF into pages stored in
↪ figures/sample-content/pdf_embed_example/split/
```

```

#
↪ pdftools::pdf_split("figures/sample-content/pdf_embed_example/Lyngs2020_FB.pdf")
#
↪ output =
↪ "figures/sample-content/pdf_embed_example/split/"

# grab the pages
pages <-
↪ list.files("figures/sample-content/pdf_embed_example/split",
↪ full.names = TRUE)

# set how wide you want the inserted PDFs to be:
# 1.0 is 100 per cent of the oxforddown PDF page width;
# you may want to make it a bit bigger
pdf_width <- 1.2

# for each PDF page, insert it nicely and
# end with a page break
cat(stringr::str_c("\\newpage \\begin{center}
↪ \\makebox[\\linewidth][c]{\\includegraphics[width=", pdf_width,
↪ "\\linewidth]{", pages, "}} \\end{center}"))

```

# ‘I Just Want to Hack Myself to Not Get Distracted’: Evaluating Design Interventions for Self-Control on Facebook

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## ABSTRACT

Beyond being the world’s largest social network, Facebook is for many also one of its greatest sources of digital distraction. For students, problematic use has been associated with negative effects on academic achievement and general wellbeing. To understand what strategies could help users regain control, we investigated how simple interventions to the Facebook UI affect behaviour and perceived control. We assigned 58 university students to one of three interventions: goal reminders, removed newsfeed, or white background (control). We logged use for 6 weeks, applied interventions in the middle weeks, and administered fortnightly surveys. Both goal reminders and removed newsfeed helped participants stay on task and avoid distraction. However, goal reminders were often annoying, and removing the newsfeed made some fear missing out on information. Our findings point to future interventions such as controls for adjusting types and amount of available information, and flexible blocking which matches individual definitions of ‘distraction’.

## Author Keywords

Facebook; problematic use; self-control; distraction; ICT non-use; addiction; focus; interruptions

## CCS Concepts

•Human-centered computing → Empirical studies in HCI;

## INTRODUCTION

Research on ‘Problematic Facebook Use’ (PFU) has investigated correlations between Facebook use and negative effects on outcomes such as level of academic achievement [35] and subjective wellbeing [58, 57]. A cross-cutting finding is that negative outcomes are associated with difficulty at exerting self-control over use, as well as specific use patterns including viewing friends’ wide-audience broadcasts rather than receiving targeted communication from strong ties [13, 58].

Much of this work has focused on self-control over Facebook use in student populations [2, 44, 46], with media multitasking research finding that students often give in to use which provides short-term ‘guilty pleasures’ over important, but aversive academic tasks [76, 88, 60]. In the present paper, we present a mixed-methods study exploring how two interventions to Facebook — goal reminders and removing the newsfeed — affect university students’ patterns of use and perceived control over Facebook use. To triangulate self-report with objective measurement, our study combined usage logging with fortnightly surveys and post-study interviews.

We found that both interventions helped participants stay on task and use Facebook more in line with their intentions. In terms of use patterns, goal reminders led to less scrolling, fewer and shorter visits, and less time on site, whereas removing the newsfeed led to less scrolling, shorter visits, and less content ‘liked’. However, goal reminders were often experienced as annoying, and removing the newsfeed made some participants fear missing out on information. After the study, participants suggested a range of design solutions to mitigate self-control struggles on Facebook, including controls for filtering or removing the newsfeed, reminders of time spent and of use goals, and removing features that drive engagement. As an exploratory study, this work should be followed by confirmatory studies to assess whether our findings replicate, and how they may generalise beyond a student population.

## RELATED WORK

### Struggles with Facebook use

Whereas many uses of Facebook offer important benefits, such as social support, rapid spread of information, or facilitation of real-world interactions [78], a substantial amount of research has focused on negative aspects [58]. For example, studies have reported correlations between patterns of Facebook use and lower academic achievement [77, 86], low self-esteem, depression and anxiety [51], feelings of isolation and loneliness [2], and general psychological distress [15]. Such ‘Problematic Facebook Use’ (PFU) has been studied under various names (including ‘Facebook dependence’ [87] and ‘Facebook addiction’ [5]), but a recent review summarised a common definition as ‘problematic behaviour characterised by addictive-like symptoms and/or self-regulation difficulties related to Facebook use leading to negative consequences in personal and social life’ [58].

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## 1.10 Including another paper in your thesis - R Markdown child document

Sometimes you want to include another paper you are currently writing as a chapter in your thesis. Above ??, we described the simplest way to do this: include the other paper as a pdf. However, in some cases you instead want to include the R Markdown source from this paper, and have it compiled within your thesis. This is a little bit more tricky, because you need to keep careful track of your file paths, but it is possible by including the paper as a child document. There are four main steps:

1. Include the paper as a child document
2. Make file paths compatible with knitting the article on its own, as well as when it's include in your thesis
3. Make header levels correct
4. Make figure widths correct

### 1.10.1 An example paper in another folder

Take this simple example (files for this are in this [GitHub repository](#)):

```
|--paper_to_include
|   |--my_paper.Rmd
|   |--data
|   |   |--cat_salt.csv
|   |--figures
|   |   |--cat.jpg
|
|--thesis
```

As the chart suggests, you have another folder, **paper\_to\_include/** living in the same containing folder as your thesis folder. In the **paper\_to\_include** folder, the file **my\_paper.Rmd** is where you write the paper. In **my\_paper.Rmd**, you read in a CSV file found in the subfolder **data/cats.csv**, and also an image from the subfolder **figures/cat.jpg**.

### 1.10.2 Step 1: Include paper as a child document

In your thesis folder, create an Rmd file for the chapter where you want to include another paper. Add one or more code chunks that include R Markdown files from that paper as child documents:

```
# Including an external chapter

```{r child = "../paper_to_include/my_paper.Rmd"}
```
```

### 1.10.3 Step 2: Make file paths compatible

Use parameters to adjust the file path of images based on values you set in the YAML header of an R Markdown file. In **my\_paper.Rmd**, create a parameter called `other_path` and set it to an empty string:

```
---
title: "A fabulous article in a different folder"
params:
  other_path: ""
---
```

In **my\_paper.Rmd**, put this at the start of the filepath when you read in data or include images:

```
library(tidyverse)
library(knitr)

cat_data <- read_csv(str_c(params$other_path, "data/cats.csv"))
include_graphics(str_c(params$other_path, "figures/cat.jpg"))
```

Finally, in your thesis folder's **index.Rmd** file, also create the parameter `other_path`. But here, set it to where the **paper\_to\_include/** folder is relative to your thesis folder:

```
params:
  other_path: "../paper_to_include/"
```

## Note on HTML output

Note that if you want to host an HTML version on your thesis online, you will need to include graphics in the content that you host online - the internet obviously won't be able to see filepaths that are just referring to stuff in another folder on your computer!

### 1.10.4 Step 3: Make sure header levels are correct

Unless the paper you want to include is also written as a book, your header levels are probably going to be off. That is, the level 1 headers (`# Some header`) you use for main sections in the other paper turns into chapter titles when included in your thesis.

To avoid this, first *increment all heading levels by one* in `paper_to_include/my_paper.Rmd` (`# Some header -> ## Some header`). Then in `paper_to_include/` create a lua filter that decrements header levels by one: Create a text file, save it as `reduce_header_level.lua`, and give it the content below.

```
function Header(el)
  if (el.level <= 1) then
    error("I don't know how to decrease the level of h1")
  end
  el.level = el.level - 1
  return el
end
```

In the YAML header of `paper_to_include/my_paper.Rmd`, use this filter:

```
---
title: "A fabulous article in a different folder"
params:
  other_path: ""
output:
  pdf_document:
    pandoc_args: ["--lua-filter=reduce_header_level.lua"]
---
```

Now, your header levels will be correct both when you knit the paper on its own and when its included in your thesis.

NOTE: There might be no need to use a lua filter to shift heading - it seems you could simply use `pandoc_args: ["--shift-heading-level-by=-1"]` (see <https://pandoc.org/MANUAL.html#reader-options>)

### 1.10.5 Step 4. Make sure figure widths are correct

It might be that your figure widths when knitting your paper on its own, and when including it in your thesis, need to be different. You can again use parameters to set figure widths.

Imagine you want figure width to be 80% of the page width when knitting your paper on its own, but 100% in your thesis. In `paper_to_include/my_paper.Rmd`, first add a parameter we could call `out_width` and set it to the string “80%”:

```
---
title: "A fabulous article in a different folder"
params:
  other_path: ""
  out_width: "80%"
output:
  pdf_document:
    pandoc_args: ["--lua-filter=reduce_header_level.lua"]
---
```

Then, make sure use that parameter to set the output width when you include figures in `paper_to_include/my_paper.Rmd`:

```
```{r, out.width=params$out_width, fig.cap="A very funny cat"}
include_graphics(str_c(params$other_path, "figures/cat.jpg"))
```
```

Finally, create the parameter `out_width` in your thesis’ `index.Rmd` file:

```
params:
  other_path: "../paper_to_include/"
  out_width: "80%"
```

Now, the output width of your figure will be 80% when knitting your paper on its own, and 100% when knitting it as child document of your thesis.

## 1.11 Customizing citations and referencing

### 1.11.1 Using a .csl file with pandoc

See section ??.

The only drawbacks to letting pandoc handle citations is that (i) it does not support chapter bibliographies, (ii) if you're a LaTeX veteran, you might be more comfortable with `biblatex` or `natbib`.

### 1.11.2 Using biblatex

To use `biblatex` to handle citations, first uncomment this in `index.Rmd`, YAML header:

```
use-biblatex: true  
bib-latex-options: "style=authoryear, sorting=nyt, backend=biber,  
↪ maxcitenames=2, useprefix, doi=true, isbn=false,  
↪ uniqueness=false"
```

Then tell R Markdown to use `biblatex` when inserting citations, by setting `citation_package: biblatex`:

```
output:  
bookdown::pdf_book:  
  citation_package: biblatex
```

To customise the appearance of citations, change `bib-latex-options`. For example, to get **numerical citations**, with references in order of their appearance in the text, set it to

```
bib-latex-options: "style=numeric-comp, sorting=none,  
↪ backend=biber, maxcitenames=2, useprefix, doi=true, isbn=false,  
↪ uniqueness=false"
```

### Adding chapter bibliographies

If you would like chapter bibliographies, first add “`refsection=chapter`” to the `biblatex` options, for example like this:

```
bib-latex-options: "refsection=chapter, style=authoryear,  
↪ sorting=nyt, backend=biber, maxcitenames=2, useprefix,  
↪ doi=true, isbn=false, uniquename=false"
```

Second, set the parameter `insertHeadingInPDF: false` in `index.Rmd`, to suppress the inclusion of a ‘References’ heading at the end of the thesis.

```
params:  
  insertHeadingInPDF: false
```

Finally insert this line at the end of each chapter, to print the bibliographies there:

```
\printbibliography[segment=\therefsection,heading=subbibliography]
```

### 1.11.3 Using natbib

To use natbib to handle citations, first uncomment this in `index.Rmd`, YAML header:

```
use-natbib: true  
natbib-citation-style: authoryear #for science, you might want  
↪ numbers,square  
natbib-bibliography-style: templates/ACM-Reference-Format.bst #e.g.  
↪ "plainnat", unsrtnat, or path to a .bst file
```

Then tell R Markdown to use natbib when inserting citations, by setting `citation_package: natbib`:

```
output:  
  bookdown::pdf_book:  
    citation_package: natbib
```

To customise the appearance of citations, change what `.bst` file you point to in `natbib-bibliography-style`.

## 1.12 Customizing the page headers and footers (PDF)

This can now be done directly in **index.Rmd**'s YAML header. If you are a LaTeX expert and need further customisation that what's currently provided, you can tweak the relevant sections of **templates/template.tex** - the relevant code is beneath the line that begins `\usepackage{fancyhdr}`.

## 1.13 Diving in to the OxThesis LaTeX template (PDF)

For LaTeX minded people, you can read through **templates/template.tex** to see which additional customisation options are available as well as **templates/ociamthesis.cls** which supplies the base class. For example, **template.tex** provides an option for master's degree submissions, which changes identifying information to candidate number and includes a word count. At the time of writing, you must set this directly in **template.tex** rather than from the YAML header in **index.Rmd**.

## 1.14 Customising to a different university

### 1.14.1 The minimal route

If the front matter in the OxThesis LaTeX template is suitable to your university, customising **oxforddown** to your needs could be as simple as putting the name of your institution and the path to your university's logo in **index.Rmd**:

```
university: University of You
university-logo: figures/your-logo-here.pdf
```

### 1.14.2 Replacing the entire title page with your required content

If you have a **.tex** file with some required front matter from your university that you want to replace the OxThesis template's title page altogether, you can provide



a filepath to this file in `index.Rmd`. `oxforddown`'s sample content includes and example of this — if you use the YAML below, your front matter will look like this:

```
alternative-title-page:
↪ front-and-back-matter/alt-title-page-example.tex
```

|                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Title of your Thesis</p> <p>John Doe</p>                                                                                                                                      | <p><b>Thesis committee</b></p> <p><b>Promotor:</b><br/>Prof.dr. J. Smith<br/>Professor of Geo-information Science and Remote Sensing<br/>Wageningen University</p> <p><b>Co-promotors:</b><br/>Dr. Name of co-promotor<br/>Assistant Professor, Laboratory of Geo-information Science and Remote Sensing<br/>Wageningen University</p> <p><b>Other members:</b><br/>Prof.dr. Jury member 1, Wageningen University<br/>Prof.dr. Jury member 2, Affiliation<br/>Prof.dr. Jury member 3, Affiliation<br/>Prof.dr. Jury member 4, Affiliation</p> <p>This research was conducted under the auspices of the C.T. de Wit Graduate School of Production Ecology &amp; Resource Conservation (PEARC)</p> | <p>Title of your thesis</p> <p>John Doe</p> <p><b>Thesis</b><br/>submitted in fulfillment of the requirements for the degree of doctor at<br/>Wageningen University<br/>by the authority of the Rector Magnificus<br/>Prof. Dr A.P.J. Mol,<br/>in the presence of the<br/>Thesis Committee appointed by the Academic Board<br/>to be defended in public<br/>on Date of your defense<br/>at 4 p.m. in the Aula</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p>John Doe<br/>Title of your thesis<br/>77 pages<br/>PhD thesis, Wageningen University, Wageningen, NL (2015)<br/>With references, with summary in English<br/>ISBN XXX-YYY</p> | <p>For Yihai Xie</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Acknowledgements</p> <p>This is where you will normally thank your advisor, colleagues, family and friends, as well as funding and institutional support. In our case, we will give our praises to the people who developed the ideas and tools that allow us to push open science a little step forward by writing plain-text, transparent, and reproducible theses in R Markdown.</p> <p>We must be grateful to John Graham for inventing the original version of Markdown, to John MacFarlane for creating Pandoc (<a href="http://pandoc.org">http://pandoc.org</a>) which converts Markdown to a large number of output formats, and to Yihai Xie for creating <code>oxfrd</code> which introduced R Markdown as a way of embedding code in Markdown documents, and <code>bookdown</code> which added tools for technical and longer-form writing.</p> <p>Special thanks to Chester Ismay, who created the <code>theisidown</code> package that helped many a PhD student write their theses in R Markdown. And a very special thanks to John MacFarlane, whose adoption of Stan Evans' adaptation of Keith Gillen's original <code>marke</code> template for writing an Oxford University DPhil thesis in LaTeX provided the template that I in turn adapted for R Markdown.</p> <p>Finally, grateful thanks to JJ Adriaens, the founder and CEO of RStudio, and Hadley Wickham, the mastermind of the <code>tidyverse</code> without whom we'd all just given up and done data science in Python instead. Thanks for making data science easier, more accessible, and more fun for us all.</p> <p>Urbk Lengua<br/>Linacre College, Oxford<br/>2 December 2018</p> |